

Chapter 6 Additional Worksheets

Probabilities & Estimation — Confidence Intervals

These additional worksheets give extra practice on sampling variability and confidence intervals, in the same style as the Chapter 1–5 worksheets. Work through the reference notes and worked example first, then attempt the exercises.

Reference: confidence intervals for a mean (used throughout this worksheet)

A **sample** is used to estimate an unknown **population** mean μ . Different samples give different sample means – this is *sampling variability*.

For a sample of size n with sample mean $\bar{x}$ and standard deviation s , the **standard error** of the mean is

$$SE = \frac{s}{\sqrt{n}}$$

A confidence interval (CI) for μ has the form

$$\bar{x} \pm z \times SE$$

where z depends on the confidence level:

Confidence level	90%	95%	98%	99%
z -value	1.64	1.96	2.33	2.58

A “95% confidence interval” means: if we repeated the sampling process many times and built a CI each time, about 95% of those intervals would contain the true population mean μ . It is **not** correct to say there is a 95% probability μ lies in this particular interval.

6.1 Constructing and interpreting confidence intervals

Whenever a survey, poll or scientific study reports a mean value, it should also report a margin of error or confidence interval – without one, the reader cannot judge how reliable the estimate is.

Activity 1

A café owner records the waiting time (in minutes) for a random sample of 40 customers. The sample mean is 4.8 minutes with a standard deviation of 1.5 minutes.

Without doing any calculation, would you expect a 99% confidence interval for the true mean waiting time to be wider or narrower than a 90% confidence interval? Explain your reasoning.

Example (worked, for reference)

A quality controller measures the mass of 50 randomly selected chocolate bars from a production line. The sample mean mass is 102.3 g with a standard deviation of 4.2 g.

a) Calculate a 95% confidence interval for the true mean mass of chocolate bars from this line.

b) Interpret this confidence interval in context.

Solution.

a) Standard error: $SE = \frac{4.2}{\sqrt{50}} = 0.594$ (3 s.f.)

95% CI: $102.3 \pm 1.96 \times 0.594 = 102.3 \pm 1.16$

So the interval is (101.1, 103.5) to 1 d.p.

b) We are 95% confident that the true mean mass of all chocolate bars produced on this line lies between 101.1 g and 103.5 g. If this sampling process were repeated many times, about 95% of the resulting intervals would contain the true population mean.

Key point. A confidence interval always has the form *estimate* $\pm$ *margin of error*. The margin of error gets **smaller** as the sample size n increases, and gets **larger** as the confidence level increases.

Exercise 6A

1. A random sample of 36 bags of sugar has mean mass 1.02 kg and standard deviation 0.06 kg.
 - a) Calculate the standard error of the mean.
 - b) Find a 90% confidence interval for the true mean mass of a bag of sugar.
 - c) Find a 99% confidence interval for the true mean mass, and compare its width with your answer to part b).
2. A school nurse measures the resting heart rate of a random sample of 64 Year 12 students. The sample mean is 72 beats per minute with a standard deviation of 8 beats per minute. Calculate a 95% confidence interval for the true mean resting heart rate of all Year 12 students at the school.
3. A random sample of 25 journey times (in minutes) for a particular bus route has mean 34.5 and standard deviation 5.0.
 - a) Calculate a 98% confidence interval for the true mean journey time.
 - b) The bus company claims the mean journey time is 32 minutes. Comment on this claim in light of your confidence interval.
4. A researcher wants a narrower confidence interval for a population mean without reducing the confidence level. State **two** ways the researcher could achieve this, and explain the effect each would have on the width of the interval.

Exercise 6B — interpreting and comparing intervals

1. Two independent samples are taken from the same population of exam scores.
 - Sample A ($n = 20$): 95% CI is (58.2, 65.8)
 - Sample B ($n = 120$): 95% CI is (60.5, 63.5)
 - a) Which sample gives the more precise estimate of the population mean? Justify your answer.
 - b) Suggest a reason why the two intervals have different widths.
2. A 90% confidence interval for a population mean is calculated as (48.1, 51.9) from a sample of size 50.
 - a) State the sample mean.
 - b) Without recalculating from scratch, explain whether a 99% confidence interval from the *same* sample would be wider or narrower than (48.1, 51.9).
3. A newspaper reports: “A survey of 30 voters found that 95% of people support the new policy.” Explain why this sentence confuses two different uses of a percentage, and rewrite it as a statistically correct statement about a confidence interval.

Review. After working through this chapter you should:

- understand why a sample mean varies from sample to sample (sampling variability)
- be able to calculate the standard error of the mean, $SE = s/\sqrt{n}$
- be able to construct a confidence interval for a population mean using $\bar{x} \pm z \times SE$
- be able to interpret a confidence interval correctly in context
- understand how sample size and confidence level affect the width of an interval